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CNN AND LSTM - BASED FORECASTING OF NDVI TRENDS FOR ENVIRONMENTAL MONITORING

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***Abstract*—This study explores the use of Long Short-Term Memory (LSTM) networks to predict vegetation dynamics from satellite data cubes. Using Sentinel-2 multispectral data, we demonstrate the feasibility of predicting vegetation indices trends for improved environmental monitoring. We also combined Convolutional Neural Networks (CNN) with bidirectional long short-term memory (Bi-LSTM) networks to extract spatio-temporal features from Normalized Difference Vegetation Index (NDVI) data cubes and modeling temporal dependencies. The proposed models achieve competitive performance with Root Mean Square Error (RMSE) values of 0.01 for the first model, and 0.17 for the second model, highlighting their potential for operational use in ecological and agricultural applications.**

***Index Terms*—LSTM, vegetation dynamics, satellite data cubes, Sentinel-2, machine learning, environmental monitoring.**

I. Introduction

Forecasting vegetation dynamics from satellite imagery has become increasingly critical for environmental monitoring and risk assessment in the context of global climate change. The ability to forecast vegetation patterns and their temporal evolution is essential for early warning systems, agricultural management, and ecosystem conservation. Satellite-based vegetation prediction enables stakeholders to anticipate and mitigate potential risks such as drought impacts, crop failures, and forest degradation before they become severe. This proactive approach is particularly valuable given the increasing frequency of extreme weather events and their impacts on vegetation health.

Previous studies have extensively explored the use of recurrent neural networks for vegetation index prediction. Early work [1] identified forest change patterns using Vegetation Indices (VI) and Markov chain techniques. The authors used Landsat images to calculate the VI corresponding to the years 2010 and 2020. In [2] the authors investigate the use of LSTM with attention mechanisms (ALSTM), to predict the dynamics of vegetation Leaf Area Index (LAI) based on meteorological variables. The study validates the performance of the ALSTM model across various biomes in North America, using satellite-derived data and flux tower observations. The results demonstrate high spatial and temporal consistency between ALSTM predictions and satellite-based LAI data, highlighting the model's potential for forecasting ecological processes in land surface models. [3] addresses the challenge of precise crop yield prediction, a key factor in ensuring global food security. Using Sentinel-2 satellite imagery, the research introduces an advanced deep learning framework that combines an attention-LSTM model with randomized Principal Component Analysis (r-PCA) to manage the temporal and spectral complexities of remote sensing data by filtering redundant information and improving the predictive accuracy of corn yield at the field scale. This study [4] uses LSTM and Gated Recurrent Units (GRU) to predict short-term VI using MODIS and Sentinel-2 data. The pixel-based (FCLSTM, FCGRU) and patch-based (ConvLSTM, ConvGRU) models were evaluated globally across regions, vegetation types, and seasons. Results show superior accuracy of RNN models (average RMSE $\approx$ 0.03) compared to traditional MLPs, with pixel-based models outperforming patch-based models in regions with higher outlier proportions. In [5] the authors predicts vegetation behavior using PlanetScope time-series data. Six models, including simple RNN, LSTM, GRU, Conv1D, RF, and ARIMA, were compared for their accuracy in forecasting NDVI. The simple RNN outperformed other methods, achieving a mean RMSE of 0.051, demonstrating

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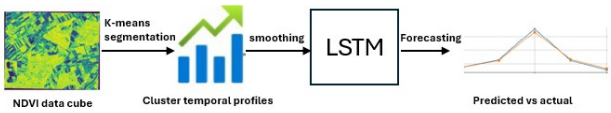


Fig. 1. LSTM-Based NDVI Prediction Workflow

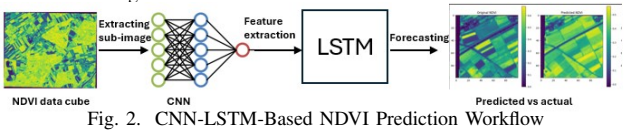
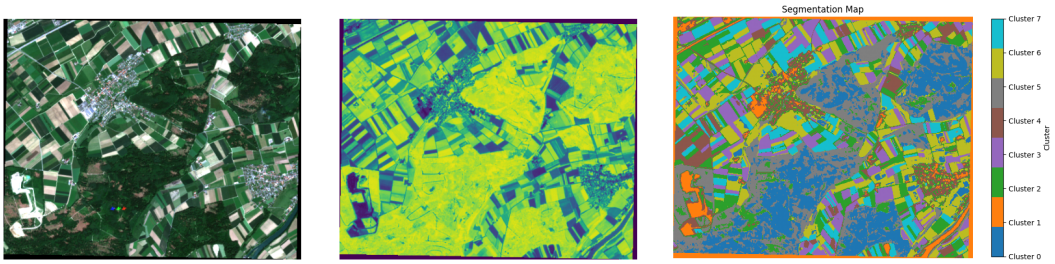


Fig. 2. CNN-LSTM-Based NDVI Prediction Workflow

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(a) (b) (c)

Fig. 3. a) Natural color Sentinel-2 scene (25.05.2018) providing contextual support for interpreting the clustering results; b) NDVI map; c) Segmentation map.

its effectiveness for high-resolution spatio-temporal vegetation predictions in Mediterranean areas. Similar studies [6] have shown that models as ConvLSTM and PredRNN, are effective in capturing spatio-temporal dependencies for vegetation index prediction.

Several recent studies have demonstrated the applicability of LSTM-based models for forecasting NDVI time series in diverse environmental contexts. For instance, in [7], LSTM was combined with fuzzy clustering to model vegetation dynamics on Lesvos Island, achieving RMSE values between 0.017 and 0.079. Similarly, [8] integrated Seasonal Trend Decomposition (STL) with BiLSTM to predict NDVI in Beijing, significantly improving performance (RMSE = 0.0209) compared to using a standard LSTM model (RMSE = 0.0385). In another case, [9] applied LSTM to forecast NDVI for a potato crop in Gujarat, India, reporting an RMSE of 0.23.

Unlike these studies, which primarily focus on temporal modeling of NDVI, our proposed approach introduces a spatio-temporal deep learning framework by combining CNN for spatial feature extraction with Bi-LSTM networks for capturing temporal dependencies. This architectural design enables the model to simultaneously learn spatial patterns within NDVI data cubes and their evolution over time—offering improved generalization and operational relevance in environmental monitoring scenarios. Our contribution lies in this dual-stage modeling framework that fuses spatial and temporal contexts within a single trainable architecture, which is rarely addressed in existing NDVI prediction studies.

II. Methodology

We considered two approaches, LSTM and CNN-LSTM. In the first one (Fig. 1), we begin by delineating clusters of temporal patterns using the K-means algorithm. For each cluster, we compute the NDVI mean. The mean value of each cluster is then smoothed using the Savitzky-Golay filter to reduce abrupt fluctuations and noise while preserving essential signal characteristics, such as peaks and subtle variations, which are crucial for detailed spectral data analysis. These smoothed values are then used as input to the LSTM model, resulting in predictions based solely on temporal patterns.

The CNN-LSTM method (Fig. 2) takes a different approach by using a CNN to extract a set of spatio-temporal features from each cluster of the time-series scene. This feature set is then utilized to predict a scene from the time series, comparing the forecasted scene with the actual image from the dataset. This method captures spatial dependencies in addition to temporal dynamics, offering a more comprehensive prediction framework.

A. Data Description

The dataset comprises 92 Sentinel-2 scenes (2018-2022) for a region in Switzerland, including vegetation indices such as NDVI. The dataset consists of 2 regions of interest, both from the same Sentinel-2 tile, showing diverse landscapes, including forested and agricultural areas.

The time series is non-uniformly sampled, indicating that the number of scenes varies across months. For instance, as illustrated in Fig. 4, the $7th$ month of 2018 contains 4 scenes, whereas the $1st$, $2nd$, and $3rd$ months lack any scenes.

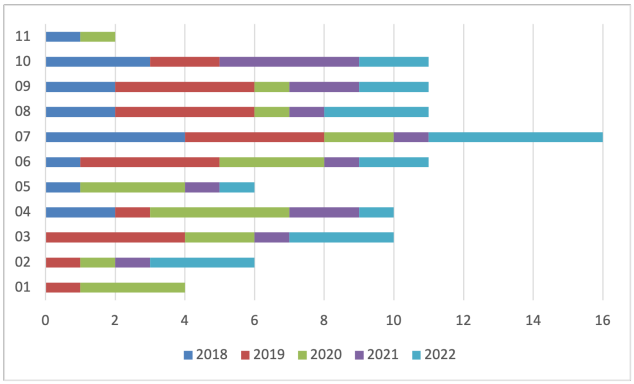


Fig. 4. Distribution of scenes over the years [10]. The dataset contains no images for the month of December across all five years. In contrast, July consistently records the highest number of usable scenes.

The data preprocessing involved resampling and constructing time-series inputs for LSTM training. The segmentation of NDVI profiles was accomplished through K-means clustering implemented on a pixel-wise basis across the time series. The initial three-dimensional NDVI data structure $\mathbf{D} \in \mathbb{R}^{h \times w \times t}$, where h represents height, w represents width, and t represents time steps, is transformed into a two-dimensional matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, where $n = h \times w$ represents the total number

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of pixels, and $p = t$ represents the temporal dimension. This transformation preserves the temporal relationships while enabling efficient clustering operations. The K-means algorithm iteratively partitions the n observations into K clusters through an optimization process. The algorithm consists of two primary steps:

Cluster Assignment Step: Each pixel's temporal profile x_i is assigned to the cluster k that minimizes the squared Euclidean distance to the cluster centroid:

$$\arg\min_k \|x_i - \mu_k\|^2$$

where μ_k represents the centroid of cluster k, and $\|\cdot\|^2$ denotes the squared Euclidean distance.

Centroid Update Step: The centroid of each cluster is recalculated as the mean of all temporal profiles assigned to that cluster:

$$\mu_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$$

where C_k represents the set of observations in cluster k, and $|C_k|$ denotes the cardinality of this set. Selecting $k = 8$ for segmentation reflects the spectral, spatial, and temporal diversity of the region. It provides sufficient granularity, allowing the model to capture both major and subtle variations within the area. Preliminary tests demonstrated that this choice produces a segmentation that improves the predictions of the LSTM model. After applying the K-means algorithm we obtained a segmentation map where each pixel in the image is associated to a specific cluster, respectively a temporal profile plot showing the characteristic NDVI patterns for each cluster along the time series. Figure 3 presents the segmentation map 3(c) alongside the true-color satellite image 3(a) and the NDVI map 3(b). The segmentation was obtained through unsupervised clustering of temporal NDVI profiles, revealing eight distinct vegetation patterns (Clusters 0–7), each represented by a unique color.

The temporal profiles of the 8 clusters correlated with the timestamps is presented in Fig. 5. Cluster 1 depicts an intense urban/industrial area while cluster 0 and cluster 5 captured various forested areas.

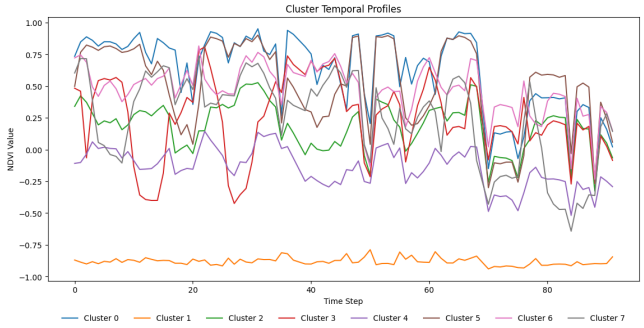


Fig. 5. Temporal profiles of the NDVI clusters.

After segmentation, a Savitzky-Golay filter was applied to smooth the data and reduce abrupt fluctuations, preserving trends and stabilizing the input for the LSTM model.

B. Models Architecture

For the LSTM approach, the LSTM network consists of two hidden layers, one with 100 units and one with 50, optimized using the Adam optimizer and Mean Squared Error (MSE) loss function. For each layer, the activation function was Rectified Linear Unit (ReLU). Since the time series of NDVI had 92 scenes, we used the last 82 values to predict the next 10.

For CNN-LSTM model the convolutional neural network consists of multiple convolutional layers, with the ReLU activation function, multiple layers of batch normalization and also Max Pooling and Dropout to prevent overfitting, since the time series has only 92 values.

C. Training and Testing

In the first scenario, the LSTM model was trained for 50 epochs, with the data split into 82 values for training and 10 values for testing. In the second scenario, the CNN utilized 80% of the data for training and 20% for testing. For the LSTM component of the model, the last 11 values were used to predict the next one. This model was trained for 90 epochs with a batch size of 4. The optimal values for the number of epochs and batch size were determined through extensive experimentation, revealing that these settings yielded the best performance.

III. Results

A. Performance Metrics

For the LSTM model, the results (the predicted values vs the actual ones from the test set) are presented in Fig. 6.

Table I summarizes the Mean Square Error (MSE) and Root Mean Square Error (RMSE) values for each cluster. As expected, the second cluster exhibited the poorest prediction performance.

For the CNN-LSTM, the results can be observed in Fig. 7. For evaluation purposes, one image was randomly selected from the temporal sequence, and multiple 100×100 pixel patches were extracted to serve as test samples. The left panel of the figure shows the true NDVI values, while the middle one represents the NDVI values predicted by the model. The spatial patterns are visually similar to the original NDVI, suggesting that the model has captured the general structure of the landscape. The right side of the figure shows a difference map (original minus predicted), highlighting the discrepancies between the true and estimated NDVI values. Positive differences (blue regions) indicate areas where the predicted NDVI is lower than the true NDVI, while negative differences (red regions) indicate overestimation by the model. The patterns of the differences shows that the model struggles with certain field boundaries or regions with abrupt transitions, due to the limited size of the dataset. Regarding the error metrics for the best case, the MSE had a value of 0.0315 and RMSE = 0.1775. The model loss over time for the best scenario is presented in Fig. 8.

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TABLE I
PERFORMANCE METRICS FOR LSTM MODEL.

Metric	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6	Cluster 7	Cluster 8
MSE	0.0002	0.0109	0.0008	0.0002	0.0013	0.0012	0.0010	0.0040
RMSE	0.0136	0.1042	0.0289	0.0141	0.0365	0.0353	0.0317	0.0543

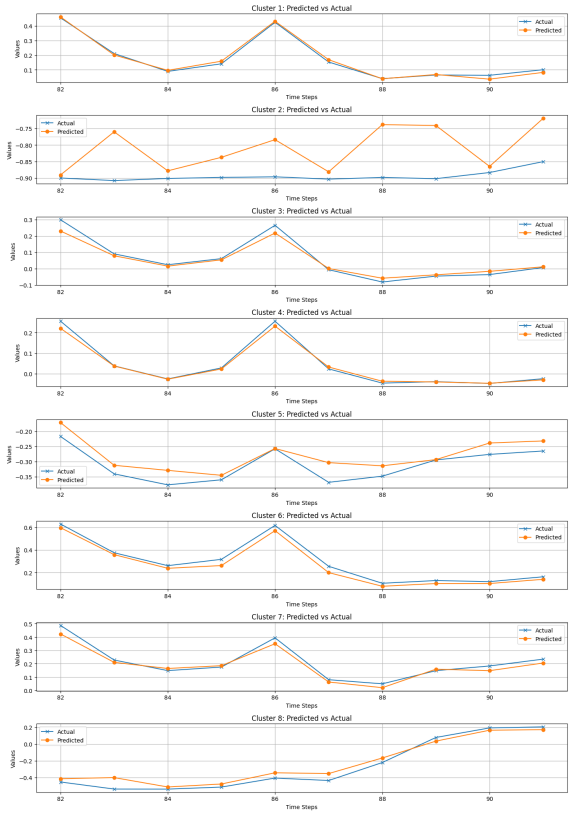


Fig. 6. Results obtained with the LSTM model. Predicted (orange) versus actual values (blue).

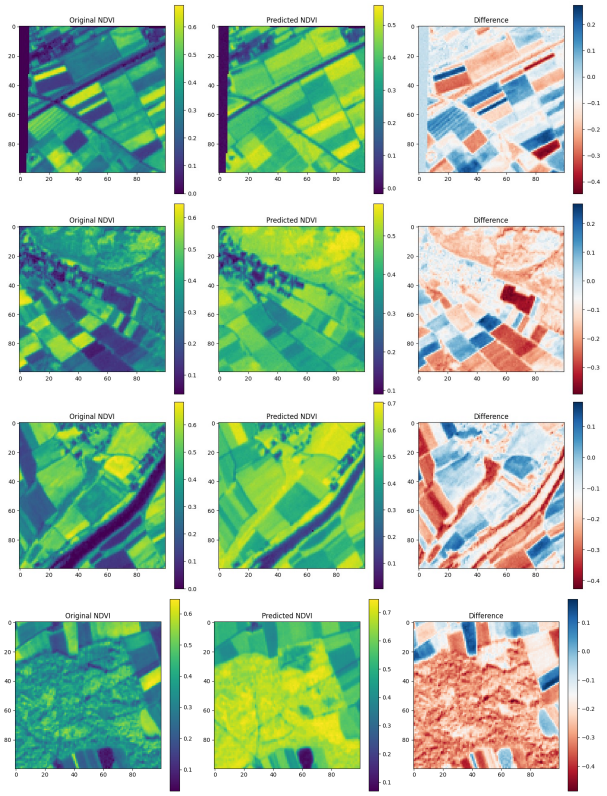


Fig. 7. Results obtained with the CNN-LSTM model. Actual (left) versus predicted image (middle) versus the difference between them (right).

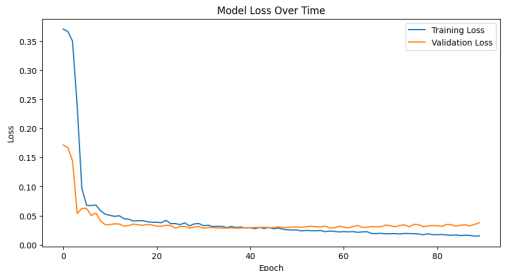


Fig. 8. Training (blue) and validation loss (orange) of the CNN-LSTM model.

IV. DISCUSSION AND CONCLUSIONS

The comparative analysis shows that LSTM outperformed CNN-LSTM in both error metrics and spatial accuracy. LSTM achieved lower MSE and RMSE across clusters, including Cluster 2, where errors were higher overall. CNN-LSTM, despite modeling spatio-temporal patterns, yielded a best-case MSE of 0.0315 and RMSE of 0.1175, higher than LSTM's top values, suggesting limited generalization due to insufficient training data. The spatial error map indicates overfitting of the CNN component to localized structures, underscoring the need for more diverse training samples. These findings highlight a key trade-off: while CNN-LSTM architectures offer enhanced spatial context integration, their performance remains sensitive to data volume and variability. In contrast, LSTM networks effectively capture temporal dependencies in vegetation indices, even under irregular sampling conditions. The size of the lookback window proved critical, with longer sequences improving trend prediction accuracy.

Overall, LSTM proved most reliable for NDVI forecasting, while CNN-LSTM shows potential in capturing spatial heterogeneity. Future work will extend to multi-pixel, multi-index predictions and integrate weather data to improve robustness.

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